

Remembering What You Don't Know: Planning with Persistent Identity Beliefs under Perceptual Aliasing

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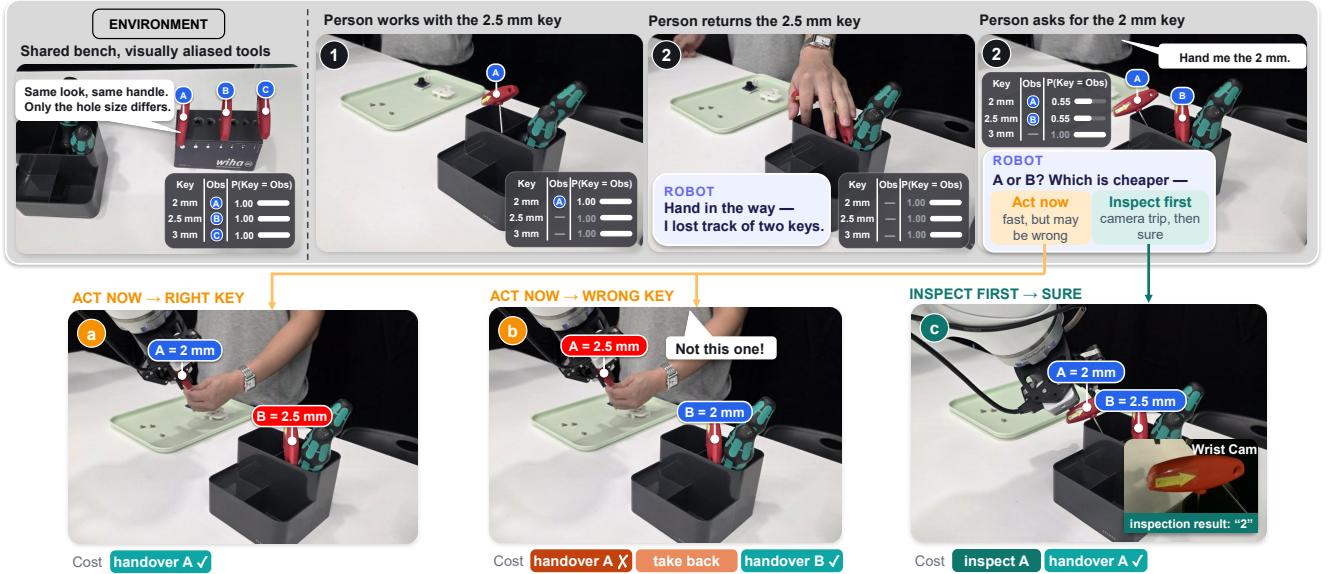


Fig. 1: Persistent identity can become ambiguous when a collaborator relocates visually aliased tools while they are unobserved or occluded. The robot maintains competing identity beliefs and chooses whether to act or inspect based on downstream cost.

Abstract—Robots that share tools with people must preserve object identity across occlusions and unobserved human manipulation, even when multiple instances are visually aliased. We present a persistent identity belief that maintains competing joint entity–observation hypotheses instead of collapsing to a single assignment, augments them with relocation uncertainty after human interaction, and exposes probabilistic identity marginals to a belief-aware planner. The planner chooses between acting on the current belief and inspecting a candidate when the expected cost of disambiguation is lower than the risk of a wrong handover. We evaluate the critical human-intervention event from a closed-loop Allen-key handover task over 99 replayed real episodes. Joint hypotheses and relocation modeling substantially reduce confidently wrong beliefs, while belief-aware planning reduces wrong handovers relative to maximum a posteriori (MAP) action and avoids the fixed overhead of always inspecting.

I. INTRODUCTION

Collaborative and assistive robots are increasingly expected to share tools with people [1], [2] by handing them over, taking them back, and reusing them later in the task.

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This work was mostly done while Brendon T. Murphy was an intern at Connecteve.

These interactions implicitly assume that the robot knows which physical tool is which. That assumption breaks when two conditions coincide (Fig. 1): multiple tools are visually aliased, sharing the same handle, texture, and geometry while differing only in a task-critical attribute (e.g., Allen keys of differing sizes); and a collaborator relocates those tools while they are unobserved. In this regime, perception can determine where visible candidates are located, but not which tool each candidate is, as current visual observations cannot distinguish them, and prior history may have been invalidated by unobserved motion.

The robot must still select an action despite identity ambiguity. Existing visuomotor memories resolve identity from the history they observed [3], [4], [5], and 3D scene graphs from appearance and geometry [6], [7], [8], [9]; neither reports identity reliability to the decision maker. Both commit to a single target, so unobserved tool displacements lead to confidently wrong handovers and costly failure recoveries. Conversely, naively verifying every candidate by inspection incurs prohibitive motion and latency overhead on every request.

Balancing action against inspection requires estimating the probability that each candidate matches the requested tool,

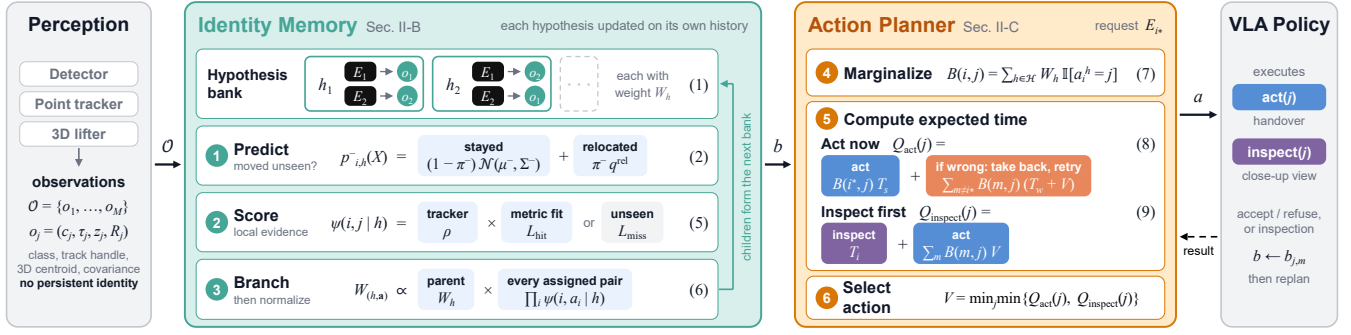


Fig. 2: System overview. Anonymous per-frame observations are fused into a joint persistent identity belief; the planner marginalizes that belief on demand for decision making, choosing between direct action and active inspection before a VLA policy executes the primitive.

and weighing the expected cost of an incorrect handover against the cost of inspection. Multi-hypothesis anchoring maintains competing identity hypotheses but remains decoupled from action selection [10], [11]. Belief-aware planning balances information acquisition against task cost, but plans over geometric and kinematic uncertainty rather than over the identity of visually aliased tools [12], [13], [14]. Also, data-association-aware variants of belief-space planning [15] focus primarily on mobile robot localization. However, these approaches primarily value sensing as a means of refining the world model, rather than directly using persistent identity uncertainty to trade the consequence of manipulating the wrong instance against the cost of resolving its identity.

To this end, we propose a memory that remembers what it does not know: a persistent identity belief paired with a belief-aware action planner. The memory maintains a bank of joint identity hypotheses that explicitly models unobserved human relocations and exposes read-only marginal identity probabilities to the planner. The planner inspects only when the expected cost of an incorrect handover exceeds the cost of inspection. We evaluate our framework on 99 replayed real episodes of unobserved tool relocation and demonstrate closed-loop execution on a physical robot. Compared to a single-hypothesis filter, our memory reduces confidently wrong beliefs from 18 to 1. Compared to always handing over the maximum a posteriori (MAP) candidate, the planner reduces wrong handovers from 9 to 2 at a 3.7 s longer mean completion time.

II. METHOD

The system operates in three stages (Fig. 2). An identity memory fuses anonymous per-frame 3D observations into a persistent joint belief, a planner selects either `act` or `inspect`, and a fine-tuned VLA executes the chosen skill. The full belief update and implementation details are provided in Appendix VI.

A. Problem Setup

We assume a closed-world episode with persistent entities $\mathcal{E} = \{E_1, \dots, E_K\}$ and $M \leq K$ current observations $\mathcal{O} = \{o_1, \dots, o_M\}$. Each $o_j = (c_j, \tau_j, z_j, R_j)$ contains a semantic class, transient track handle, 3D centroid, and stereo

measurement covariance, but no persistent identity. Observations are therefore anonymous within a visually aliased class. For a requested target entity E_{i^*} , the robot chooses between `act(j)` and `inspect(j)` over candidate observations o_j to minimize expected execution time. We omit the time index t except across frames.

B. Hypothesis-Conditioned Identity Memory

The memory maintains a discrete bank $\mathcal{H}$ of mutually consistent hypotheses. Each hypothesis $h \in \mathcal{H}$ carries a weight W_h , with $\sum_{h \in \mathcal{H}} W_h = 1$, and the state

$$h = (\mathbf{a}^h, \{\mu_{i,h}, \Sigma_{i,h}, \pi_{i,h}\}_{i=1}^K), \quad (1)$$

where $a_i^h = j$ ($a_i^h \in \mathbf{a}^h$) binds E_i to o_j (0 denotes unobserved), $(\mu_{i,h}, \Sigma_{i,h})$ are the mean and covariance of the entity's 3D position X , and $\pi_{i,h}$ is the probability that it was relocated while unobserved. The full state, including the stored track handle and the broad density used after invalidation, is given in Table III of appendix VI-A. Localized entities are represented by $\mathcal{N}(X; \mu_{i,h}, \Sigma_{i,h})$ and propagate across elapsed time Δt via a constant-position model: $\mu_{i,h}^- = \mu_{i,h}$ and $\Sigma_{i,h}^- = \Sigma_{i,h} + Q_i \Delta t$, with spatial variance capped at $\sigma_{\max}^2$ to prevent unbounded diffusion during sustained occlusions.

When an entity's semantic class is manipulated unseen, tracking for that class is suspended. On resume, a displacement transition hazard p_{reloc} updates the relocation probability: $\pi_{i,h}^- = \min(\pi_{i,h} + (1 - \pi_{i,h})p_{\text{reloc}}, \pi_{\max})$, where $\pi_{\max} < 1$ ensures the nominal state remains recoverable. The relocation alternative $q_{i,h}^{\text{rel}}(X)$ modulates the physically valid workspace support with a local Gaussian centered at $\mu_{i,h}$. The pre-observation predictive mixture is thus:

$$p_{i,h}^-(X) = (1 - \pi_{i,h}^-)\mathcal{N}(X; \mu_{i,h}^-, \Sigma_{i,h}^-) + \pi_{i,h}^- q_{i,h}^{\text{rel}}(X). \quad (2)$$

Let $p_D(X; C_s)$ be the probability that an object at X produces a valid observation under sensing context C_s . For candidate observation o_j , detection and miss likelihoods are

defined as:

$$L_{\text{hit}}(i, j | h) = \int p_D(X; C_s) \mathcal{N}(z_j; X, R_j) p_{i,h}^-(X) dX, \quad (3)$$

$$L_{\text{miss}}(i | h) = \int [1 - p_D(X; C_s)] p_{i,h}^-(X) dX. \quad (4)$$

L_{hit} scores an observed candidate using metric location and visibility, while L_{miss} penalizes an unobserved entity only to the extent that its predicted state should have been detectable.

We combine these with the hypothesis-specific 2D tracker continuation prior ρ to synthesize the local evidence:

$$\psi(i, j | h) = \begin{cases} \rho(i \rightarrow j | h) L_{\text{hit}}(i, j | h), & j > 0, \\ L_{\text{miss}}(i | h), & j = 0. \end{cases} \quad (5)$$

Thus $\psi(i, j | h)$ measures how strongly parent hypothesis h supports assigning entity E_i to observation o_j .

To propagate the belief, each parent h branches over valid one-to-one, class-consistent assignments **a**. Candidate children are scored and globally normalized as

$$W_{(h,\mathbf{a})} \propto W_h \prod_i \psi(i, a_i | h), \quad \sum_{h,\mathbf{a}} W_{(h,\mathbf{a})} = 1. \quad (6)$$

Since every valid closed-world assignment contains exactly M hit terms and $K - M$ miss terms, all hypotheses have matching density dimensions, avoiding unit-induced bias during normalization. Each child updates its geometric state according to **a**, and children with active relocation uncertainty are split into nominal and relocated branches.

C. Belief-Aware Action Planner

The planner consumes the normalized bank directly as $b = \{(h, W_h)\}_{h \in \mathcal{H}}$. For decision making, it derives the read-only marginal identity belief:

$$B(i, j) = \sum_{h \in \mathcal{H}} W_h \mathbb{I}[a_i^h = j], \quad (7)$$

which represents the marginal probability that persistent entity E_i corresponds to candidate o_j . These marginals are never fed back into tracking and filtering, preventing static scenes from self-amplifying hallucinations. Let $b_{j,m} = b(\cdot | o_j = E_m)$ denote the joint belief conditioned on candidate o_j being entity E_m ; because hypotheses encode joint assignments, this conditioning also updates the identities of the remaining candidates.

Operational costs model physical execution time. For successful handover duration T_s , wrong handover penalty T_w , and inspection duration T_i from end-effector pose e :

$$Q_{\text{act}}(j) = B(i^*, j) T_s(j, e) + \sum_{m \neq i^*} B(m, j) [T_w(j, e) + V(b_{j,m}, e_{\text{return}})], \quad (8)$$

$$Q_{\text{inspect}}(j) = T_i(j, e) + \sum_m B(m, j) V(b_{j,m}, e_{\text{inspect}}). \quad (9)$$

The value function is defined recursively as $V(b, e) = \min_j \min\{Q_{\text{act}}(j), Q_{\text{inspect}}(j)\}$ over a finite horizon. The

planner executes the minimum-cost first action and replans after inspection or human feedback.

III. EXPERIMENTS

We ask three questions: (i) does the memory produce identity beliefs that are accurate without becoming spuriously certain, (ii) which components are responsible for that behavior, and (iii) how does the resulting uncertainty change the act-or-inspect decision?

A. Evaluation Scenario and Protocol

We evaluate the identity-ambiguity event in the closed-loop interaction of Fig. 1. An original Allen key is observed in the shared bucket, after which the collaborator returns a visually similar persistent key and may displace the original while both are unobserved. The robot must then identify the original among the visible candidates. We repeat this human-intervention segment for 100 real episodes with original displacement of 0.1–8.7 cm (median 4.0 cm) and annotate the original’s identity; one episode yields no valid decision record under the observation hold, leaving 99 trials. Rather than evaluating an artificial offline proxy, this protocol isolates and controls the identical perceptual ambiguity event that occurs during the full closed-loop interaction.

The system runs on a KUKA LBR Med 14 with a wrist-mounted D405. RF-DETR [16], Track-On2 [17], Fast-FoundationStereo [18], and VP-VLA [19] implement the detector, tracker, 3D lifting, and action policy in Fig. 2. All memory variants are replayed on the same recorded observations and robot poses. Planner outcomes use the annotated identity, with failed handovers and inspections revealing the selected candidate’s identity. Action costs are forecast from demonstrations and evaluated with durations measured from real VLA rollouts.

For belief quality we report MAP identity accuracy, negative log-likelihood (NLL), Brier score, and confidently wrong (CW) trials, defined as a wrong MAP identity with confidence at least 0.9. For task behavior we report completion time, wrong handovers, and inspections. We compare the full memory with component ablations and compare our planner against MAP action, always-inspect, and an identity oracle. Additional details on the hardware setup, action costs, and evaluation protocol are provided in Appendix VII.

B. Identity Belief Quality

Table I shows that preserving joint hypotheses matters primarily for calibration. A single-hypothesis filter reduces MAP accuracy from 0.91 to 0.82, but more importantly produces 18 confidently wrong trials instead of 1. Once the wrong association is committed, the planner receives a nearly deterministic but incorrect belief and has little reason to inspect.

Relocation modeling changes confidence even when top-1 accuracy does not. Removing the relocation alternative leaves MAP accuracy at 0.91, yet NLL nearly doubles and CW rises from 1 to 9: the filter remains certain exactly where an unobserved human interaction may have invalidated

TABLE I: Identity-belief ablation on 99 replayed human-intervention episodes.

Memory	MAP $\uparrow$	NLL $\downarrow$	CW $\downarrow$
Full	0.91	0.33	1
Single hypothesis	0.82	1.26	18
w/o relocation	0.91	0.63	9
Always relocated	0.91	0.45	0
Uniform relocation only	0.79	0.42	2
w/o negative spatial update	0.68	0.87	17

TABLE II: Planner behavior by displacement of the original tool (full table in Table V). Ours decides with forecast costs; times are scored with measured durations.

Disp.	Ours		MAP action	
	Time	Wrong	Time	Wrong
All	28.46	2	24.74	9
< 3 cm	21.37	0	21.37	0
3–5 cm	27.31	0	21.37	0
5–9 cm	36.35	2	32.14	9

its remembered location. At the opposite extreme, always discarding position memory removes confident errors but loses useful history and causes more inspections. Using only a uniform relocation prior also hurts accuracy, indicating that the local component acts as a useful scene prior that moved tools are often returned near their previous location. Removing negative spatial conditioning is more severe, reducing MAP accuracy to 0.68 and producing 17 confidently wrong trials. Together, these results support the intended behavior: preserve useful metric history while retaining uncertainty when that history may no longer be valid.

C. Acting on the Belief

On the full memory, our planner makes 2 wrong handovers and 37 inspections, compared with 9 wrong handovers for MAP action and 99 inspections for always-inspect. MAP action is 3.7 ± 1.0 s faster on average scored with measured costs as the memory’s MAP candidate is already correct in 91% of trials and a rejected handover is relatively cheap in this setup. The planner therefore does not dominate MAP on mean time in this particular cost regime; its benefit is the ability to trade execution risk against verification cost rather than committing to either extreme.

The displacement breakdown makes this behavior clearer. Below 3 cm, the belief is near-certain and our planner acts directly in every trial, matching MAP action and the oracle. In the 5–9 cm regime, where relocation creates the strongest identity ambiguity, ours reduces wrong handovers from 9 to 2 at a cost of 4.2 s per request (36.4 vs. 32.1 s). The overall time gap comes mainly from 3–5 cm trials, where the underconfident memory and an underestimated inspection cost (Sec. VII-D) trigger 14 inspections that retrospectively prove unnecessary.

Fig. 3 makes this trade-off explicit. Holding the recorded beliefs fixed, we vary the inspection and wrong-handover costs used by the planner’s decision and charge every resulting policy with the measured durations. The operating points form a front from MAP action to zero wrong handovers that lies well below random inspection: removing 7 of the 9 wrong handovers takes 26 inspections and 1.8 s per request

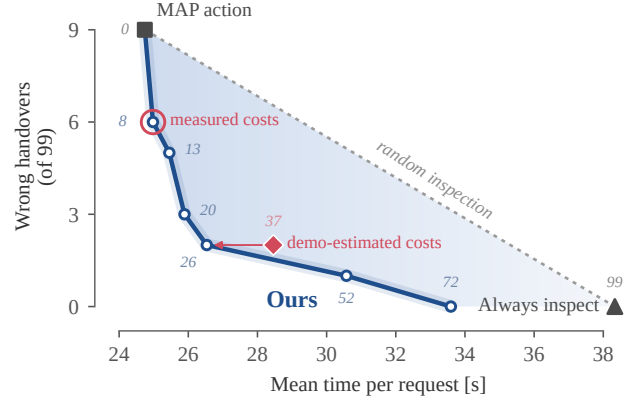


Fig. 3: Time–error trade-off on the full memory. Each point is the planner deciding with different costs, scored with measured costs. Circle/diamond: deciding with measured/forecast costs (Table II). Dotted: random mix of MAP action and always-inspect. Numbers: inspections out of 99.

over MAP actions, where random inspection would need 77, and zero wrong handovers are reached 4.7 s faster than always-inspect. Deciding with the measured durations places the planner on this front; the demonstration-derived forecast used in Table II lands 1.9 s off it at the same error count. The same belief thus serves tasks with different failure penalties, given accurate action costs.

D. Closed-Loop Demonstration and Discussion

We additionally execute the complete interaction of Fig. 1 end to end. The collaborator uses and returns the 2 mm key and then the 2.5 mm key to the same bucket, mixing the two visually aliased tools. When the 2 mm key is requested again, the robot faces the ambiguity quantified above: a confident belief leads to direct handover, while an ambiguous one can trigger inspection first (Appendix VII-E). The evaluated event is thus embedded in the full memory–planning–action loop rather than being only an offline association problem.

The results expose two distinct calibration requirements. Identity beliefs must avoid overconfident errors that suppress useful inspection and underconfidence that adds needless ones, while the planner needs a reasonable action-cost model to avoid inspecting too often. Neither substitutes for the other: cost calibration cannot repair a collapsed identity belief, and belief calibration alone cannot determine whether information gathering is worth its execution time.

IV. CONCLUSION

We presented a persistent identity belief that preserves competing object identities under perceptual aliasing and unobserved relocation, together with a belief-aware planner that trades acting against inspection based on downstream task cost. Real-robot replay experiments show that joint hypotheses and relocation modeling reduce confidently wrong beliefs and wrong handovers. Although the limited evaluation does not fully demonstrate the planner’s recursive capability, the results support exposing persistent identity uncertainty directly from the world model to planning. Future work will

incorporate priors over human interaction, object relocation, and task sequence to improve generalization across tasks and environments.

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APPENDIX

V. RELATED WORK

We organize prior work by five questions. For memory: what is kept when object identity is ambiguous, whether a change that happens while the object is unobserved is represented, and what is passed on to action selection. For planning: what criterion triggers additional sensing, and which quantity is uncertain.

A. Object Identity in Robot Memory

Memory in visuomotor policies. Embodied-SlotSSM propagates per-object slots with a slot state-space model and is evaluated on LIBERO-Mem, a benchmark that contains visually identical objects [3]. Chameleon recalls traces of past observation tokens to resolve decisions that the current frame leaves aliased [4], Keyframe-Chaining VLA retrieves selected keyframes and follows one of three identical cubes through a swap [5], and MemoryVLA retrieves perceptual and cognitive tokens from a memory bank [20]. In these tasks the robot observes the event that determines identity, and the policy outputs one action without an estimate of how reliable the remembered identity is. POT-VLA is closer to our setting: it keeps role-indexed 3D object records whose confidence and visibility fields can mark a record as uncertain and trigger re-observation [21], although by rule and without alternatives over which instance is which.

Scene graphs and spatio-temporal maps. ConceptGraphs associates detections with map objects greedily by visual-feature similarity and point-cloud overlap [7]. LOST-3DSG matches objects by text attributes followed by a spatial-consistency check, handles objects that move while unobserved, and sets aside objects with identity conflicts as uncertain [8]. PUF replaces hard association gates with per-observation association probabilities and accumulates class and relation evidence in Dirichlet distributions, but computes these probabilities per frame and does not carry alternative associations across time [6]. Khronos detects when objects appear or disappear, yet associates object fragments only geometrically, so that, as its authors note, fragments that have moved are not associated [9]. None of these maintains joint identity hypotheses over time or uses identity uncertainty to choose an action.

Multi-hypothesis anchoring. Multiple hypothesis tracking defers association decisions by maintaining alternative association hypotheses, each a set of compatible tracks, and pruning them over time [22]. PMHA brings this into object anchoring: it keeps a capped set of world hypotheses with probabilities, filters each object’s position with several behavior models, one of which spreads the object uniformly over the table and gains weight while the object is unobserved, and uses a visibility probability so that missed detections count as evidence [10]. DAA adds a knowledge base of expected spatial relations to re-associate lost anchors, and evaluates two identical boxes whose positions are exchanged while the camera is occluded [11]. These are the closest precedents for our memory; joint hypotheses and a moved-while-unobserved alternative are both present in PMHA. The difference lies in what leaves the memory. PMHA reports the most probable hypothesis and DAA a single MAP assignment, so the alternatives are not used to decide which object to act on. DAA further separates identical instances through relational context, which is unavailable when the instances offer no such cue. A follow-up on the same world model schedules re-observations by trading expected information gain in object position against travel cost, and refines an object’s position by a fixed rule before every grasp [23]; it does not weigh the cost of acting on a wrong identity.

B. Deciding When to Sense

Belief-space task and motion planning. POMDPs are the standard model for manipulation under partial observability [24]. Kaelbling and Lozano-Pérez plan in belief space so that look actions follow from goals on the belief, and explicitly assume that perceived object identities are correct [12]. Garrett et al. plan cost-sensitively over particle beliefs of object poses, inflating the cost of an action by its probability of failure, for a set of unique, known objects [13]. STRUG rewards the reduction of task-relevant uncertainty and is demonstrated on the kinematic models of articulated objects [14]. Our act-or-inspect rule follows the same principle, sensing when it lowers expected cost, and differs in the uncertain quantity: which visible instance is the requested one.

Association-aware planning. DA-BSP represents the belief as a mixture over data-association hypotheses and adds an ambiguity term to the planning cost, which enables active disambiguation [15]. It is the closest formal precedent for letting association hypotheses drive sensing, and it is formulated and evaluated for robot localization in perceptually aliased environments rather than for choosing which instance to manipulate.

Gathering information for a decision or for a map. Decision region determination selects tests until all remaining hypotheses fall into a single decision region, so information is gathered only as far as the decision requires [25]. It assumes noise-free tests and never acts under residual risk, whereas our planner may hand over a tool whose identity is not certain. Manipulation-enhanced mapping instead selects views and pushes to maximize expected information

gain over a metric-semantic map [26], an objective that is independent of any downstream decision.

Foundation-model planners. VAP-TAMP verifies action preconditions and effects with a VLM, moves the camera when the VLM judges the view insufficient, and corrects a single scene graph when an object has been displaced [27]. STAR lets a VLM choose between querying an observation memory and acting in the scene to retrieve objects whose location may have changed [28]. Both decide to look by model judgment or rule rather than by comparing costs, and both keep a single estimate of the world.

C. Positioning

Joint identity hypotheses with a moved-while-unobserved alternative exist in anchoring [10], and cost-sensitive sensing exists in belief-space planning [13]. Our contribution is to connect the two for the object being manipulated: the memory exposes identity probabilities derived from its hypothesis bank instead of a single estimate, and the planner uses them to compare the expected cost of acting with that of inspecting.

VI. FULL BELIEF UPDATE AND PLANNING DETAILS

This appendix provides the recursive update loop underlying the hypothesis-conditioned identity memory and the belief-aware action planner. The temporal frame index t is omitted where operations are scoped within an instantaneous sensing cycle; the persistent hypothesis index h is maintained as a subscript ($\mu_{i,h}$ and $\Sigma_{i,h}$ for entity E_i under hypothesis h). We formalize transitions between three spatial representations: (i) a localized metric Gaussian, (ii) a broad non-parametric support density on invalidation, and (iii) a latent relocation alternative maintained compactly within parent hypotheses.

A. Notation and State Variables

Table III summarizes the mathematical symbols and state representations used throughout the memory and planner modules.

a) Relocation Density with Local Memory Prior: An object left in the workspace may be picked up and placed elsewhere, or merely nudged while a collaborator handles its neighbors. To cover both cases, the entity-specific relocation density $q_{i,h}^{\text{rel}}(X)$ mixes the class background density $q_c^{\text{rel}}(X)$, which accounts for arbitrary re-placement, with a Gaussian centered at the remembered centroid $\mu_{i,h}$, which accounts for small displacements:

$$q_{i,h}^{\text{rel}}(X) \propto q_c^{\text{rel}}(X) \left[(1-w) + w \frac{\mathcal{N}(X; \mu_{i,h}, \sigma_{\text{loc}}^2 I)}{\mathcal{U}(X; \Omega_{\text{valid}})} \right] \quad (10)$$

with $w = 0.4$ and $\sigma_{\text{loc}} = 5$ cm. Normalization by $\mathcal{U}(X; \Omega_{\text{valid}}) = 1/|\Omega_{\text{valid}}|$ yields a dimensionless likelihood ratio. Because $q_c^{\text{rel}}(X)$ assimilates negative visibility evidence across views, this local mode naturally dissolves once the robot inspects the remembered coordinate and observes empty space.

B. Observation Lifting and Gating

a) Median Centroid and Anisotropic Stereo Noise: For each detection mask belonging to an active track, unmasked pixels with valid stereoscopic depth are projected into world coordinates, forming point cluster $\mathcal{P}_j = \{p_l\}_{l=1}^{N_j} \subset \mathbb{R}^3$ after statistical depth outlier removal. The 3D centroid $z_j \in \mathbb{R}^3$ is the coordinate-wise median:

$$z_j = \text{median}(\mathcal{P}_j) \quad (11)$$

Centroid measurement covariance $R_j \in \mathbb{S}_{++}^3$ models triangulation geometry along the camera viewing ray $\hat{r}_j = (z_j - p_{\text{cam}})/\|z_j - p_{\text{cam}}\|$:

$$R_j = (1 + f_{\text{out}}) \left[\frac{1}{N_{\text{eff}}} \text{cov}(\mathcal{P}_j) + \sigma_{\text{floor}}^2 I \right] + \sigma_Z^2 \hat{r}_j \hat{r}_j^\top \quad (12)$$

where $N_{\text{eff}} = \min(N_j, 16)$ caps variance reduction against correlated disparity errors, f_{out} is the fraction of rejected depth outliers, $\sigma_{\text{floor}}^2 I$ provides baseline calibration tolerance, and $\sigma_Z = Z^2 \sigma_d / (f b_{\text{stereo}})$ represents triangulation noise at depth Z for focal length f , baseline b_{stereo} , and subpixel disparity noise σ_d .

b) Temporal Observation Hold: To prevent border artifacts and initialization instability from corrupting tracking, newborn tracks must maintain spatial stability within 1 cm across $F_{\text{hold}} = 5$ consecutive frames before entering the active observation set $\mathcal{O}$. Detections whose 2D masks intersect image boundaries are held immediately. Held detections are quarantined ($o_j \notin \mathcal{O}$), and detection probability p_D is set to zero within their projected masks to prevent false empty-space carving.

c) Spatial Merge and Closed-World Gating: Same-class observations whose 3D centroids lie within 1 cm are merged to rectify split masks. Under the closed-world assumption, at most K_c detections are retained per semantic class c .

C. Observability Field and Miss Likelihoods

a) Continuous Observability Field: Given sensing context C_s (camera pose, intrinsics, depth bounds), continuous detection probability $p_D(X; C_s)$ at coordinate $X \in \mathbb{R}^3$ factorizes as:

$$p_D(X; C_s) = p_{\text{det}} \mathbb{I}_{\text{FOV}}(X; C_s) \mathbb{I}_{\text{depth}}(X; C_s) \gamma(X; C_s) \quad (13)$$

clipped to $[10^{-3}, 1 - 10^{-3}]$ within the frustum and zero outside. $\mathbb{I}_{\text{FOV}}$ tests image bounds, $\mathbb{I}_{\text{depth}}$ enforces stereoscopic range $Z_{\text{cam}} \in [0.05 \text{ m}, 0.6 \text{ m}]$, and $\gamma(X; C_s)$ accounts for occlusions (0.1 across gripper finger masks, 0.0 inside held-back detection masks, and 1.0 elsewhere). Baseline recall is set to $p_{\text{det}} = 0.5$.

b) Expected Detectability and Miss Likelihood: For predicted spatial distribution $p_{i,h}^-(X)$, the expected detectability is:

$$d_{i,h} = \int_{\Omega_{\text{valid}}} p_D(X; C_s) p_{i,h}^-(X) dX \quad (14)$$

TABLE III: Notation used by the persistent identity belief and planner.

Symbol	Meaning
$\mathcal{E} = \{E_i\}_{i=1}^K$	Fixed registry of K persistent entities in the closed-world episode.
$\mathcal{O} = \{o_j\}_{j=1}^M$	Retained observations at the current frame. Observation $o_j = (c_j, \tau_j, z_j, R_j)$ contains semantic class, 2D track handle, 3D centroid, and measurement covariance.
$h \in \mathcal{H}$	Global world hypothesis $h = (\mathbf{a}^h, \{\mu_{i,h}, \Sigma_{i,h}, r_{i,h}, q_{i,h}, \pi_{i,h}\}_{i=1}^K)$ maintaining association vector $\mathbf{a}^h$ alongside metric centroids $\mu_{i,h}$, covariances $\Sigma_{i,h}$, track references $r_{i,h}$, broad densities $q_{i,h}$, and latent relocation probabilities $\pi_{i,h}$.
W_h	Normalized probability mass of hypothesis h , with $\sum_h W_h = 1$.
a_i^h	Assignment of entity E_i under hypothesis h ($a_i^h = j > 0$ assigns to o_j ; $a_i^h = 0$ indicates unobserved).
$(\mu_{i,h}, \Sigma_{i,h})$	Mean and covariance of the localized Gaussian state.
$r_{i,h}$	Stored track handle and age under hypothesis h , used strictly for tracker continuation evidence.
$q_{i,h}(X)$	Non-parametric broad density over Ω_{valid} , initialized upon memory invalidation.
$q_c^{\text{rel}}(X)$	Shared class-level relocation density conditioned on negative views since interaction ended.
$q_{i,h}^{\text{rel}}(X)$	Entity-specific relocation density modulating $q_c^{\text{rel}}(X)$ with local memory around $\mu_{i,h}$.
$\pi_{i,h}$	Latent relocation probability for E_i , strictly bounded in $[0, \pi_{\text{max}}]$ to preserve filter recoverability.
Ω_{valid}	Bounded geometric support of the table surface and static fixtures.
C_s	Sensing context (camera extrinsics, intrinsics, depth range, gripper mask, held-back track masks).
$p_D(X; C_s)$	Probability that an object at X is detected under sensing context C_s .
$\rho(i \rightarrow j h)$	Hypothesis-conditioned tracker-continuation prior.
$B(i, j)$	Read-only marginal identity belief $P(E_i = o_j)$ derived from the hypothesis bank.
$p_i(X)$	Read-only marginalized spatial density of entity E_i .
b	Discrete planner belief distribution $\{(h, W_h)\}_{h \in \mathcal{H}}$.
$b_{j,m}$	Planner belief conditioned on observation candidate o_j matching persistent entity E_m .

The complementary unassigned miss likelihood is:

$$L_{\text{miss}}(i | h) = 1 - d_{i,h} = \int_{\Omega_{\text{valid}}} [1 - p_D(X; C_s)] p_{i,h}^-(X) dX \quad (15)$$

For localized Gaussians, Eq. (15) is evaluated via 7-point unscented cubature along principal covariance axes. For broad density $q_{i,h}^-(X)$ with grid cells $\{x_k, w_k^-\}_{k=1}^{N_q}$, it reduces to discrete summation:

$$L_{\text{miss}}(i | h) \approx \sum_{k=1}^{N_q} w_k^- [1 - p_D(x_k; C_s)] \quad (16)$$

D. Hit Likelihood, Tracker Continuation, and Local Evidence Synthesis

a) *Hypothesis-Conditioned Tracker Prior*: The 2D continuation prior $\rho(i \rightarrow j | h)$ relates stored reference handle $r_{i,h}$ to observation handle τ_j . If $r_{i,h} = \tau_j$ within age $\Delta t_{\text{track}} \leq 30$ frames:

$$\alpha = \min(\alpha_{\min} + (\alpha_{\max} - \alpha_{\min})v_j, 1 - \epsilon_{\text{track}}) \quad (17)$$

with visible point ratio $v_j \in [0, 1]$, $\alpha_{\min} = 0.6$, $\alpha_{\max} = 0.98$, and margin $\epsilon_{\text{track}} = 0.02$. Remaining same-class observations share residual mass evenly:

$$\rho(i \rightarrow j' | h) = \frac{1 - \alpha}{M_{c_i} - 1}, \quad \forall j' \in \mathcal{O}_{c_i} \setminus \{j\} \quad (18)$$

If $r_{i,h}$ is unbound or stale ($\Delta t_{\text{track}} > 30$), continuation reverts to the uniform baseline $\rho(i \rightarrow j | h) = 1/M_{c_i}$.

b) *Metric Hit Likelihoods*: For localized Gaussians, the hit likelihood is evaluated directly at observation centroid z_j to avoid image-border integration artifacts:

$$L_{\text{hit}}^{\text{loc}}(i, j | h) = p_D(z_j; C_s) \mathcal{N}(z_j; \mu_{i,h}^-, \Sigma_{i,h}^- + R_j) \quad (19)$$

For broad densities $q_{i,h}^-(X)$, continuous spatial convolution over quadrature cells evaluates as:

$$L_{\text{hit}}^{\text{broad}}(i, j | h) \approx \sum_{k=1}^{N_q} w_k^- p_D(x_k; C_s) \mathcal{N}(z_j; x_k, R_j + \Sigma_{\text{cell}}) \quad (20)$$

where $\Sigma_{\text{cell}} = (\Delta^2/12)I_3$ accounts for cubic cell dispersion ($\Delta = 3$ cm).

c) *Local Evidence Synthesis*: Local association factor $\psi(i, j | h)$ combines tracking continuity with geometric likelihoods:

$$\psi(i, j | h) = \begin{cases} \rho(i \rightarrow j | h) L_{\text{hit}}(i, j | h), & j > 0, \\ L_{\text{miss}}(i | h), & j = 0 \end{cases} \quad (21)$$

E. Global Data Association and Hypothesis Bank Management

a) *Admissible Assignment Space*: Entities $\mathcal{E}$ and observations $\mathcal{O}$ decouple by semantic class. The joint assignment space $\mathcal{A}$ factorizes as:

$$\mathcal{A} = \bigtimes_{c=1}^C \mathcal{A}_c, \quad |\mathcal{A}| = \prod_{c=1}^C \frac{K_c!}{(K_c - M_c)!} \quad (22)$$

where $\mathbf{a}_c \in \mathcal{A}_c$ enforces bijective entity-to-observation pairing.

b) *Child Weighting and Global Normalization*: Each parent hypothesis h branches over valid assignments $\mathbf{a} \in \mathcal{A}$. Child weights accumulate in the log domain:

$$\log \widetilde{W}_{(h,\mathbf{a})} = \log W_h + \sum_{i=1}^K \log \psi(i, a_i | h) \quad (23)$$

Global normalization sums over all candidate children across all parents:

$$W_{(h,\mathbf{a})} = \frac{\widetilde{W}_{(h,\mathbf{a})}}{\sum_{h' \in \mathcal{H}_{\text{prior}}} \sum_{\mathbf{a}' \in \mathcal{A}} \widetilde{W}_{(h',\mathbf{a}')}} \quad (24)$$

Because every valid assignment pairs exactly M entities and leaves $K - M$ unassigned, continuous density dimensions (m^{-3M}) cancel identically in Eq. (24).

c) Bank Maintenance: To constrain hypothesis growth, the bank applies three sequential passes:

- 1) **Duplicate Merging:** Candidates with identical discrete assignments and spatial centroids within $\|\mu_{i,(h,a)} - \mu_{i,(h',a')}\| \leq 1$ mm across all entities are merged by weight summation.
- 2) **Cumulative Mass Pruning:** Candidates with $W < \epsilon_h = 10^{-6}$ are pruned, retaining the top set covering 0.9999 cumulative mass.
- 3) **Capacity Truncation:** The bank is capped to $H_{\max} = 16$ hypotheses ($H \in [8, 64]$) and renormalized.

F. Posterior State Correction

a) Localized Kalman Updates: For localized Gaussian states assigned to observation o_j ($a_i = j > 0$), innovation covariance is $S_{ij,h} = \Sigma_{i,h}^- + R_j$ and Kalman gain is $K_{ij,h} = \Sigma_{i,h}^- S_{ij,h}^{-1}$:

$$\mu_{i,(h,a)}^+ = \mu_{i,h}^- + K_{ij,h}(z_j - \mu_{i,h}^-) \quad (25)$$

$$\Sigma_{i,(h,a)}^+ = (I - K_{ij,h})\Sigma_{i,h}^- \quad (26)$$

with $\Sigma_{i,(h,a)}^+$ symmetrized and projected onto $\mathbb{S}_{++}^3$. The entity binds to handle τ_j , resets track age $\Delta t_{\text{trk}} \leftarrow 0$, and clears relocation uncertainty ($\pi \leftarrow 0$). If another unassigned entity held τ_j , it forfeits the handle.

b) Broad State Collapse: When an observation is assigned to an unlocalized broad entity ($q_{i,h}^-(X)$), the smoothly varying prior acts as a local constant across the sharp measurement covariance R_j . The posterior collapses directly onto $\mathcal{N}(z_j, R_j)$, materializing as a localized Gaussian with $\mu^+ = z_j$ and $\Sigma^+ = R_j$.

c) Spatial Carving on Misses: When an entity produces no observation ($a_i = 0$), broad grid weights update pointwise via negative Bayesian conditioning:

$$w_k^+ = \frac{w_k^- [1 - p_D(x_k; C_s)]}{\sum_{l=1}^{N_q} w_l^- [1 - p_D(x_l; C_s)]} \quad (27)$$

Class background density $q_c^{\text{rel}}(X)$ is conditioned identically over incoming views. Localized Gaussians are projected into camera coordinates, truncated against the frustum, and moment-matched to unimodal parameters.

d) Lazy Relocation Splits: For entities carrying relocation uncertainty ($\pi_{i,h}^- > 0$), state splitting is deferred until observations require disambiguation. If unassigned ($a_i = 0$), the bank does not branch; $\pi_{i,h}$ updates in place via Bayes' rule:

$$\pi_{i,(h,a)}^+ = \frac{\pi_{i,h}^- L_{\text{reloc,miss}}(i | h)}{(1 - \pi_{i,h}^-) L_{\text{nom,miss}}(i | h) + \pi_{i,h}^- L_{\text{reloc,miss}}(i | h)} \quad (28)$$

If assigned ($a_i = j > 0$), the parent splits lazily into nominal and relocated children weighted respectively by:

$$\Psi_{\text{nom}} = (1 - \pi_{i,h}^-) \rho(i \rightarrow j | h) L_{\text{hit}}^{\text{loc}}(i, j | h) \quad (29)$$

$$\Psi_{\text{reloc}} = \pi_{i,h}^- \rho(i \rightarrow j | h) L_{\text{hit}}^{\text{broad}}(i, j | h) \Big|_{q=q_{i,h}^{\text{rel}}} \quad (30)$$

In both instantiated child hypotheses, unresolved relocation probability resets to zero ($\pi \leftarrow 0$).

G. Belief-Aware Action Planner

a) Belief Representation and Conditioning: The downstream planner operates directly over discrete hypothesis belief $b = \{(h, W_h)\}_{h \in \mathcal{H}}$. Marginal identity association is queried as $B(i, j) = \sum_{h \in \mathcal{H}} W_h \mathbb{I}[a_i^h = j]$. Hypothetical inspection of observation o_j revealing entity E_m yields conditional belief $b_{j,m}$ by projecting hypothesis weights proportionally to $W_h \mathbb{I}[i_h(j) = m]$, where $i_h(j)$ is the entity assigned to o_j under hypothesis h . On hardware execution, inspection reading y is scored by vision-language likelihood $P(y | E_m) = \epsilon_y + (1 - \epsilon_y) s_m(y)$, with $s_m(y)$ the vision-language model's normalized score of the answer for E_m and $\epsilon_y = 0.02$. Human handovers condition belief deterministically: acceptance confirms the target entity, while refusal fixes the non-target identity.

b) Expected-Cost Action Value Functions: Let $T_s(j, e)$ denote successful handover duration from end-effector pose e to candidate z_j , $T_w(j, e)$ denote failed handover duration including refusal pause and return to $e_{\text{return}}(j)$, and $T_i(j, e)$ denote inspection duration to $e_{\text{inspect}}(j)$. For requested target E_{i^*} :

$$Q_{\text{act}}(j; b, e) = B(i^*, j) T_s(j, e) + \sum_{m \neq i^*} B(m, j) [T_w(j, e) + V(b_{j,m}, e_{\text{return}}(j))] \quad (31)$$

$$Q_{\text{inspect}}(j; b, e) = T_i(j, e) + \sum_{m=1}^K B(m, j) V(b_{j,m}, e_{\text{inspect}}(j)) \quad (32)$$

The optimal policy minimizes expected duration:

$$V(b, e) = \min_{j \in \mathcal{J}(b)} \{Q_{\text{act}}(j; b, e), Q_{\text{inspect}}(j; b, e)\} \quad (33)$$

Value function $V(b, e)$ is evaluated recursively to a finite horizon of 8 actions with dynamic programming memoization over tuples (b, e) . At the horizon limit, search terminates with a greedy fallback to the highest-probability candidate. Replanning executes receding-horizon from the updated belief.

H. Action Policy and Visual Prompting

We use VP-VLA as the low-level policy for executing the planner-selected manipulation primitive on visually aliased tools. The planner actions in Eqs. (31)–(32) are parameterized by a candidate observation o_j . Each observation already contains a localized 3D centroid z_j , obtained by lifting the detected point cluster into the world frame as described in Sec. VI-B and Eq. (11). When the candidate is incorporated into a localized entity state, this measurement also updates

the corresponding hypothesis-conditioned position through Eqs. (25) and (26).

VP-VLA System 2 specifies the target object through a crosshair-style visual prompt. For the candidate o_j selected by the planner, we project its current 3D centroid z_j into the hand-eye-calibrated wrist-camera image and continuously update the crosshair during execution. Let cT_W denote the calibrated transformation from the world frame to the current wrist-camera frame and $\Pi(\cdot)$ the camera projection function. The prompt location is therefore

$$u_j = \Pi\left({}^cT_W \begin{bmatrix} z_j \\ 1 \end{bmatrix}\right), \quad (34)$$

where u_j is the image coordinate at which the crosshair is overlaid. Reprojection is repeated as the wrist camera moves, so the visual prompt remains registered to the selected physical candidate throughout the primitive.

For direct handover and inspection, the accompanying natural-language instruction is given in the form “{act, inspect} the highlighted object.” A third, refusal-recovery primitive is invoked after a rejected handover and is trained separately from the handover and inspection skills.

We collect 260 demonstrations in total across three independently trained skill primitives—handover, inspection, and refusal-recovery—with approximately 50–100 demonstrations per skill, and fine-tune VP-VLA on the combined dataset. To encourage the policy to rely on the visual prompt rather than on scene-level shortcuts, we additionally collect contrastive demonstrations in which the same visual scene is paired with different highlighted target objects.

VII. ADDITIONAL EXPERIMENTAL ANALYSIS

A. Evaluation Scenario, System, and Protocol

The quantitative benchmark isolates the same human-intervention event that occurs inside the end-to-end handover sequence of Fig. 1. Before the intervention, the robot has observed the persistent identity and position of an original Allen key. While the wrist camera is directed elsewhere, the collaborator returns another visually similar persistent key to the shared workspace and may displace the original. When the camera returns, both candidates can be localized geometrically but their persistent identities may be ambiguous. We repeat this post-intervention recovery event for 100 real episodes. The original is displaced by 0.1–8.7 cm (median 4.0 cm); one episode produces no valid decision record under the observation hold, leaving 99 trials. The final image of each episode is annotated with the original key identity.

The physical setup is shown in Fig. 4. Using a wrist-mounted camera that provides the egocentric sensing stream for the proposed system, RF-DETR provides tool detections, Track-On2 supplies short-term point-track continuity, and Fast-FoundationStereo provides stereo depth for lifting detections into 3D. The external camera shown in Fig. 4 is used only as the external-image input to VP-VLA. It is never used by the persistent identity memory or by the action planner; both operate exclusively on the partial observations obtained from the wrist-mounted camera. Thus, the identity belief in

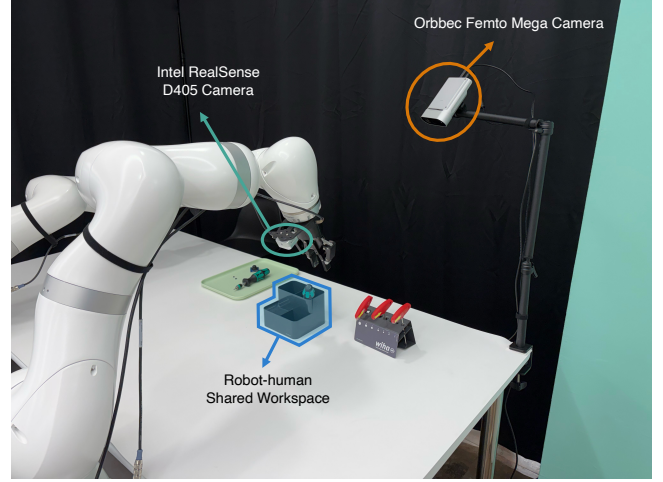


Fig. 4: Real-world experimental setup. A KUKA LBR Med 14 R820 with a Robotiq 2F-85 gripper operates over the shared robot-human workspace. The wrist-mounted Intel RealSense D405 provides the egocentric observations used by the persistent identity memory and planner. The external Orbbec Femto Mega provides an additional third-person image only to VP-VLA during skill execution and is not accessible to the memory or planner.

Sec. VI and the planner belief in Sec. VI-G do not have access to a third-person view, even though the downstream policy receives the additional external image required by VP-VLA.

The evaluation contains a recorded component and a counterfactual policy component. Camera frames and robot poses are recorded on the real robot and replayed through every memory variant, so all variants receive identical sensory evidence. At the decision point, policy outcomes are computed against the annotated identity: a wrong candidate is treated as refused and identified by the collaborator, while an inspection reveals the candidate identity. Episode-level completion time is therefore counterfactual, but each primitive cost is grounded in real robot execution. Ten deployed VLA rollouts per skill give mean durations of 21.37 s for a successful direct handover, 16.96 s additional time for inspection, and 15.73 s for refusal/recovery before the next attempt.

The deployed planner’s internal forecast is calibrated from the demonstration data gathered to train the downstream VLA policy, with corresponding costs of 20.82 s, 11.24 s, and 14.53 s. Because this forecast substantially underestimates inspection, we report the effect of this mismatch separately in Sec. VII-D. The belief metrics are evaluated independently of either cost model.

B. Detailed Analysis of Identity Memory

Table IV reports the complete ablation, including both belief quality and the downstream behavior induced by the same expected-cost decision rule. The results distinguish three properties that are easy to conflate: top-1 identity accuracy, calibration of the full belief, and downstream decision utility.

TABLE IV: Full memory ablation on the replayed human-intervention episodes. CW denotes confidently wrong trials: an incorrect MAP identity with confidence at least 0.9.

	Memory	MAP $\uparrow$	NLL $\downarrow$	Brier $\downarrow$	CW $\downarrow$	Time [s] $\downarrow$	Wrong $\downarrow$	Inspect $\downarrow$
	Full	0.91	0.33	0.10	1	28.46	2	37
Hypotheses	Single hypothesis	0.82	1.26	0.18	18	28.12	18	0
Relocation	w/o relocation	0.91	0.63	0.09	9	24.74	9	0
	Always relocated	0.91	0.45	0.13	0	30.48	1	51
	Uniform relocation only	0.79	0.42	0.14	2	30.34	2	48
Evidence	w/o negative spatial update	0.68	0.87	0.25	17	33.17	26	12
	w/o observation hold [†]	0.59	0.85	0.27	6	35.32	28	21

[†] Without observation hold, all 100 recorded episodes yield a valid decision; all other rows use the common 99-episode subset.

TABLE V: Policies acting on the full memory, overall and by displacement of the original. Times are reported as mean $\pm$ standard error of the mean (SEM) across episodes and are computed from measured robot primitive durations.

Displacement	n	Ours			MAP action		Always inspect	
		Time [s]	Wrong	Inspect	Time [s]	Wrong	Time [s]	Inspect
All	99	28.46 $\pm$ 0.93	2	37	24.74 $\pm$ 1.08	9	38.33 $\pm$ 0.00	99
< 1 cm	10	21.37	0	0	21.37	0	38.33	10
1–3 cm	18	21.37	0	0	21.37	0	38.33	18
3–5 cm	40	27.31	0	14	21.37	0	38.33	40
5–9 cm	31	36.35	2	23	32.14	9	38.33	31

Joint hypotheses prevent confident identity collapse. Replacing the bank with a single surviving hypothesis lowers MAP accuracy from 0.91 to 0.82, but the larger effect is on calibration: NLL increases from 0.33 to 1.26 and confidently wrong trials increase from 1 to 18. Once an early association error is committed, later observations are interpreted only through that history, so the planner receives a nearly deterministic but incorrect identity belief. Consistent with this failure mode, the expected-cost rule never inspects under this variant and produces 18 wrong handovers. This result motivates retaining mutually exclusive identity explanations rather than using only the current MAP association.

Relocation uncertainty matters even when MAP accuracy is unchanged. Removing the relocation alternative leaves MAP accuracy at 0.91, yet NLL nearly doubles to 0.63 and confidently wrong trials rise from 1 to 9. The filter therefore often identifies the same top candidate as the full model, but is too certain that the remembered metric state remained valid across the unobserved human interaction. This distinction is important for planning: the overconfident variant suppresses inspection and degenerates to the MAP-action behavior, producing 9 wrong handovers. At the opposite extreme, treating every post-interaction entity as relocated removes confident errors but discards useful position memory, increasing inspections from 37 to 51. The full model lies between these extremes by preserving nominal spatial memory while retaining a relocation alternative when the interaction may have invalidated it.

The relocation prior should retain local scene structure. Using only the uniform component of the relocation density lowers MAP accuracy to 0.79 and causes 48 inspections. The local Gaussian component therefore contributes more than numerical convenience: it encodes the scene prior that a tool is often only nudged from its remembered location. In this dataset that prior is informative since the collaborator typically displaces the original by only a few centimeters, while still allowing the broad component to explain

larger relocations. This prior is deliberately simple and task-dependent; richer interaction priors are a natural extension rather than a requirement of the hypothesis-bank formulation. **Negative spatial evidence must change the state, not only its weight.** Without the negative spatial update, a miss can reweight a hypothesis but leaves its spatial density unchanged. Subsequent misses are then repeatedly evaluated against the same stale spatial prior, effectively reusing evidence without removing already-observed empty regions. This produces the largest core-memory degradation: MAP accuracy falls to 0.68, NLL rises to 0.87, and 17 trials become confidently wrong, leading to 26 wrong handovers. The full method instead conditions the hypothesis-specific spatial state on the current observability field after a miss, as detailed in Sec. VI-F.

Observation holding is an implementation safeguard rather than a memory primitive. Removing the temporal observation hold yields the lowest MAP accuracy (0.59). Newly emerging or image-edge-truncated masks can have unstable 3D centroids; admitting them immediately lets transient measurements claim persistent identities before their geometry settles. We therefore view the hold as a perception-interface safeguard that prevents unreliable measurements from entering the probabilistic update, rather than as part of the core persistent-identity representation.

Overall, the ablations show why top-1 accuracy alone is insufficient for evaluating a planning-facing memory. The full model and the no-relocation variant have identical MAP accuracy, yet their confidently wrong counts and downstream behavior differ substantially. What matters to the planner is not only whether the most likely identity is correct, but whether the belief remains appropriately uncertain when the interaction history no longer determines a unique assignment.

C. Planner Behavior by Ambiguity Level

Table V groups the policy comparison by how far the original tool was displaced during the unobserved intervention.

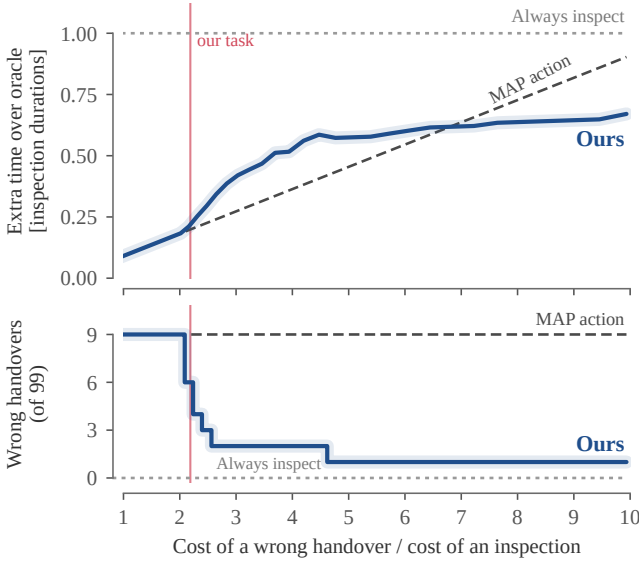


Fig. 5: Cost-ratio sweep on the full memory, with the same costs used for the decision and for scoring. The ratio is varied through the inspection duration. Top: extra time per request over the oracle, in units of one inspection. Bottom: wrong handovers. The vertical line marks the measured costs of our task.

Displacement is not itself used as a planner threshold; it serves as an interpretable proxy for how strongly the current observations conflict with the remembered spatial state.

Below 3 cm, the full memory remains near-certain and the planner acts directly in every trial, matching MAP action and avoiding the fixed cost of always inspecting. The 3–5 cm regime exposes a weakness of the current belief: it is underconfident, with MAP action correct in all 40 trials. Combined with a forecast that underestimates inspection time (Sec. VII-D), this yields 14 unnecessary inspections, which account for 2.4 s of the 3.7 s mean-time gap to MAP action. With the measured costs the planner makes only 8 inspections in total, so at least 6 of these checks would not occur without the forecast error.

The 5–9 cm regime is the setting in which the identity history is most strongly disrupted. Here the planner performs 23 inspections and reduces wrong handovers from 9 to 2, at a cost of 4.2 s per request relative to MAP action (36.35 vs. 32.14 s). Thus the planner’s useful behavior is not to inspect whenever uncertainty exists, but to concentrate verification in the region where the belief and the task cost make a wrong commitment consequential. Across all trials, always-inspect eliminates uncertainty at a large fixed overhead, whereas MAP action eliminates that overhead by accepting every identity guess. The proposed rule interpolates between these extremes using the same belief.

D. Sensitivity to Action-Cost Calibration

The act-or-inspect decision depends on both identity uncertainty and the relative execution costs. This dependence is intentional, as the same belief should rationally induce different actions in a task where inspection is cheap than

in one where a wrong handover is expensive. The cost-ratio sweep in Fig. 5 holds the recorded beliefs fixed and varies the inspection duration, so that the cost of a wrong handover relative to an inspection ranges from 1 to 10. MAP action pays no verification cost but degrades linearly as failures become more expensive, whereas always-inspect pays a constant overhead. Below a ratio of about 2, the planner acts directly in every trial and matches MAP action. Above it, the planner starts inspecting: its wrong handovers fall from 9 to 2 by a ratio of 2.7 and to 1 beyond 4.8. In time, however, MAP action remains faster up to a ratio of about 6.8, and the planner becomes preferable only beyond it. The reason is the underconfidence noted in Sec. VII-C. Trials with MAP confidence between 0.6 and 0.8 are correct in 29 of 31 cases, so inspecting them is rarely worthwhile, but the planner takes these probabilities at face value. Better-calibrated beliefs would move this crossover to lower ratios.

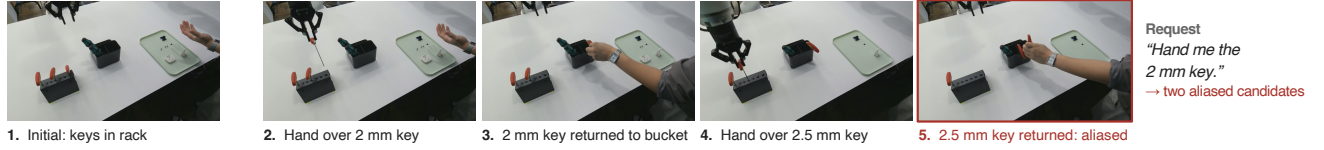
Cost-model error is therefore distinct from belief error. The demonstration-derived forecast used by the deployed planner predicts 20.82 s for a successful handover, 11.24 s for inspection, and 14.53 s for refusal/recovery, while measured deployed-policy durations are 21.37 s, 16.96 s, and 15.73 s. The main mismatch is inspection, underestimated by about 6 s. The results in Tables IV and V use the forecast for the decision and the measured durations for scoring: the planner inspects in 37 of 99 trials and takes 28.46 s per request. Deciding with the measured durations instead reduces the inspections to 8 and the time to 24.98 s, within 0.3 s of MAP action, at 6 rather than 2 wrong handovers. In Fig. 3, this measured-cost decision lies on the planner’s trade-off front, whereas the forecast-cost decision is dominated by it: the front reaches 2 wrong handovers at 26.53 s. The 1.9 s by which the forecast-cost decision misses the front is a cost-model error rather than a belief error: with the same beliefs, an accurate cost model reaches the same error count 1.9 s faster. The belief’s own contribution is the underconfidence of Sec. VII-C, which determines where the front lies.

E. Closed-Loop Demonstration

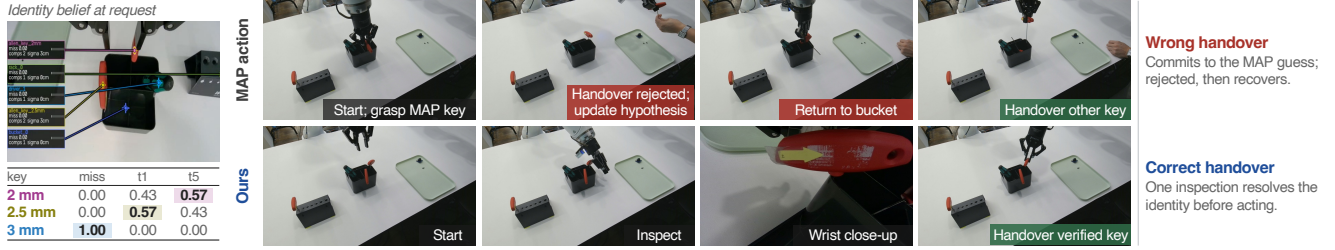
The replay benchmark isolates the same post-intervention ambiguity event that occurs naturally in the full robot interaction. To verify integration, we additionally execute the complete sequence shown in Fig. 6. Panel (a) shows the shared task prefix. Three Allen keys begin in the rack. The collaborator requests the 2 mm key, uses it, and returns it to the bucket. During the next request, the robot hands over the 2.5 mm key while observing that the 2 mm key is in the bucket. The collaborator then returns the 2.5 mm key to the same bucket and mixes the two visually aliased tools. When the 2 mm key is requested again, the robot encounters the same type of ambiguous two-candidate state used in the statistical evaluation.

Panels (b)–(c) show two representative outcomes after that shared prefix. In the high-ambiguity case, the identity posterior is close enough that MAP action commits to the wrong key and must recover after refusal; our planner instead values the remaining identity risk above the inspection cost, moves

(a) Shared task prefix deterministic handover and return; the collaborator then mixes two aliased Allen keys



(b) High ambiguity



(c) Low ambiguity

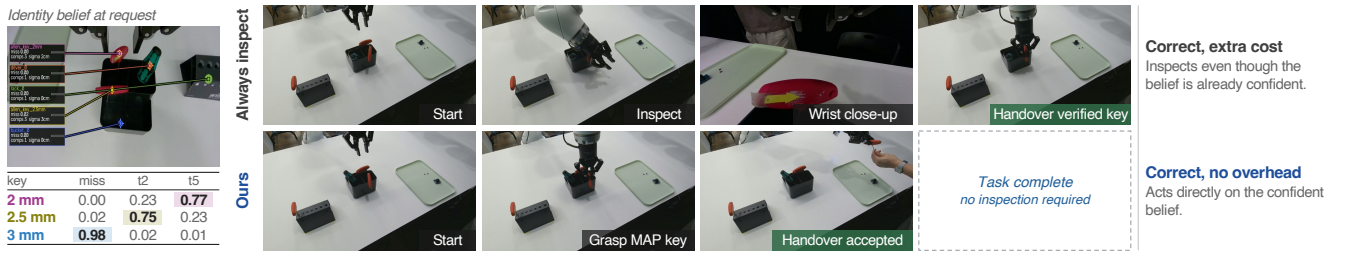


Fig. 6: Closed-loop real-robot demonstration. (a) Shared task prefix: the robot hands over the 2 mm and 2.5 mm Allen keys, which the collaborator returns to the bucket so that two visually aliased candidates remain when the 2 mm key is requested again. (b) High-ambiguity case: MAP action commits to the wrong candidate and requires refusal and recovery, whereas our planner first performs a wrist-camera close-up inspection and then hands over the verified key. (c) Low-ambiguity case: always-inspect incurs an unnecessary verification motion, while our planner acts directly on the confident identity belief. The examples illustrate how the same persistent identity posterior changes the manipulation strategy rather than serving only as an offline association estimate.

the wrist camera for a close-up observation, and then grasps the verified candidate. In the low-ambiguity case, the posterior already favors the correct candidate strongly enough that our planner acts directly, whereas always-inspect performs the same close-up verification and pays its fixed overhead. These demonstrations show that the posterior changes the executed manipulation strategy rather than serving only as an offline association score. They are intended as end-to-end system validation rather than an additional statistical benchmark; the quantitative claims are based on the repeated replay evaluation above.

F. Limitations

Value of recursive planning. In the evaluated scenario, a single inspection resolves the identities of both candidates, so the expected-cost decision reduces to a one-step comparison between acting and inspecting. On our recorded beliefs, a confidence-threshold rule that inspects whenever the MAP confidence falls below a tuned value traces nearly the same front as Fig. 3 (within 0.4 s at equal error counts). We therefore do not claim an advantage of the recursive expected-cost search over thresholding in this setting. Demonstrating it requires scenarios with more candidates, inspections that can fail, or candidate-dependent motion costs, which we leave

to our future work. What the cost-based rule provides here is an operating point derived from the action costs rather than a hand-tuned threshold; with the current underconfident beliefs, however, a threshold tuned once at our task costs remains competitive across the cost range of Fig. 5.

Belief calibration. The memory is underconfident at intermediate confidence, which, together with the underestimated inspection cost of the deployed forecast, contributes to the unnecessary inspections in the 3–5 cm regime, and delays the cost ratio at which the planner outperforms MAP action.

Sample size and scope. The evaluation consists of 99 trials of a single scenario: two visually aliased Allen keys in one bucket, one collaborator, and displacements below 9 cm. Wrong-handover counts are in the single digits, so differences of one or two between variants or operating points are within sampling noise, and we draw conclusions only from the overall pattern of the results. Policy outcomes are counterfactual, computed from the replayed beliefs and measured primitive durations rather than executed end to end for every trial. The durations come from ten successful rollouts per skill, excluding failed executions, and the local relocation prior is chosen for the displacement range of this scenario.